POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2018
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable

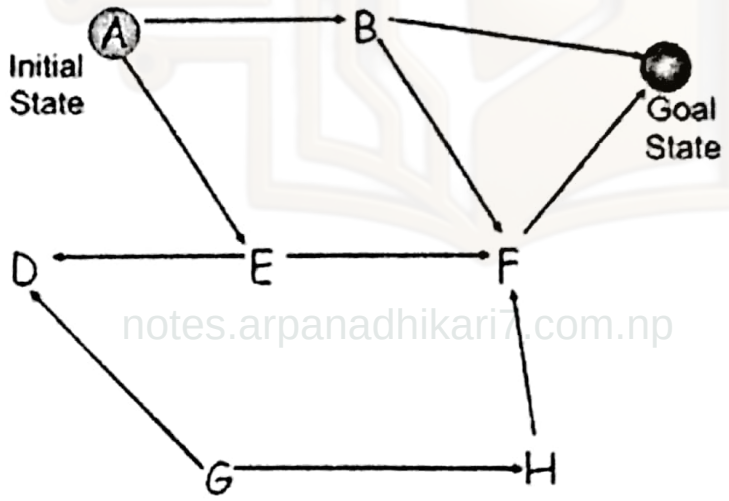
The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is Artificial Intelligence? Mention the advantages of Artificial Intelligence. 7

   b) Define an artificial agent. Specify the task environment: - performance measure, environment, actuators and sensor for a taxi driver agent and a part-picking robot agent. 8

2. a) What is a state space of a problem? For the given problem represented in graph, where initial state is "*A*" and goal state is "*C*", find all the possible state space. 8



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   b) Define Un-informed search. Explain Iterative deepening search with an example. 7

3. a) How are informed search methods more efficient than uninformed search? Explain the algorithm of A* search with a suitable example. 8

   b) What is knowledge? Describe the different approached of knowledge representation in brief. 7

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4. a) Consider the following facts:
   i. Lucy is a professor
   ii. All professors are people.
   iii. John is the dean.
   iv. Deans are professors.
   v. All professors consider the dean a friend or don't know him.
   vi. Everyone is a friend of someone.
   vii. People only criticize people that are not their friends.
   viii. Lucy criticized John.

   Now, using resolution find the answer of "Is John no friend of Lucy?"

   b) What is inductive learning? Explain with example.

5. a) Contrast between supervised and unsupervised learning. Describe the K-means learning method with example.

   b) What is deep learning? Explain in detail.

6. a) Explain a rule based expert system architecture? Mention its advantages and disadvantages.

   b) Why NLP is Difficult? Explain steps in the process of NLP.

7. Write short notes on: (**Any Two**)

   a) Clustering

   b) Machine vision

   c) Resolution refutation system

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